Smart Inventory & Supply Chain Intelligence System

An AI-powered inventory management system built with Java, MySQL & Swing — engineered for real-time intelligence, demand forecasting, and supplier evaluation.

Team

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Stack

Java 17 · MySQL · Swing

Scope

2nd Year CS Project

The Problem with Traditional Inventory

Most businesses still rely on fragmented, reactive systems — leaving them blind to demand shifts, supplier failures, and mounting losses.

No Real-Time Visibility

Stock levels update only after manual audits — too slow for fast-moving supply chains.

Zero Demand Forecasting

Businesses over-order or under-stock with no data-driven insight into future demand.

Weak Supplier Evaluation

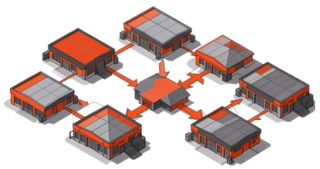
No structured scoring leads to poor vendor choices, delays, and quality failures.

Dead Stock & Excel Chaos

Unsold inventory piles up silently while teams struggle with error-prone spreadsheets.

Solution Overview

● WHAT WE BUILT



Multi-Warehouse Tracking

Unified real-time stock visibility across multiple warehouse locations from a single dashboard.

AI Demand Forecasting

Moving average algorithm trained on real-world data predicts next-month demand with seasonal adjustments.

Supplier Scoring Engine

Automated 0–100 scoring ranks suppliers by reliability, defect rate, and cost — updated continuously.

Real-Time Dashboard

Swing-based GUI delivers live KPIs, alerts, and inventory intelligence in one clean interface.

Technology Stack

Java 17

Core language — OOP principles, generics, collections, and streams.

Java Swing

GUI framework powering the real-time dashboard and data tables.

MySQL + JDBC

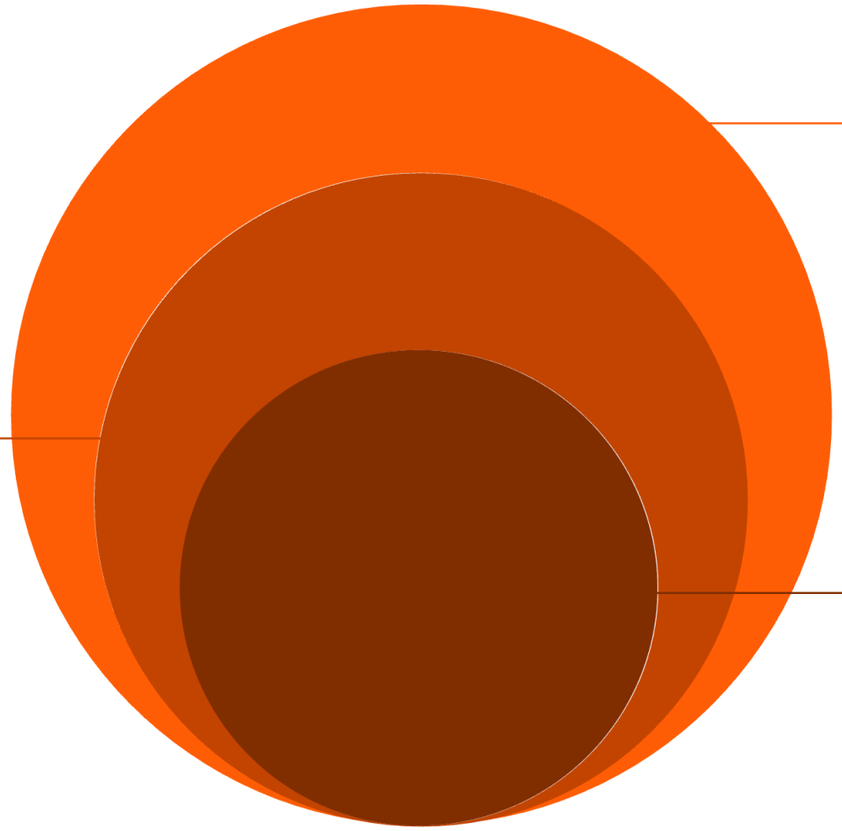
Relational data persistence with connection pooling via JDBC.

Maven + Libraries

Build automation with JFreeChart, OpenCSV, and iTextPDF integrations.

System Architecture

A clean 3-layer architecture separates concerns — keeping the GUI, business logic, and data access independently testable and maintainable.



Data Access

DAO pattern with MySQL via JDBC

Business Logic

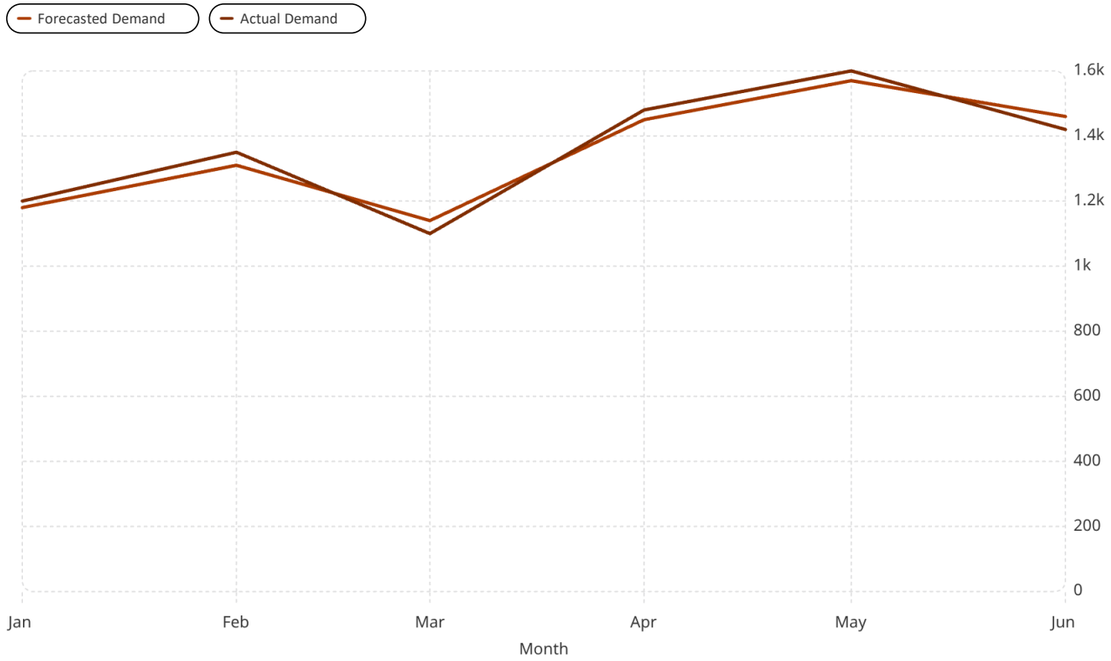
Service classes: forecasting, scoring, rebalancing

Presentation

Java Swing GUI: dashboards, forms, alerts

Each layer communicates strictly downward — the GUI never touches the database directly, ensuring modularity and production-grade separation of concerns.

AI-Powered Demand Forecasting



How It Works

The forecasting engine is trained on **73,000 real transaction records** and applies a **3-month moving average** with seasonal adjustment factors.

Output: predicted next-month demand per SKU, enabling proactive reordering before stockouts occur.

~14%

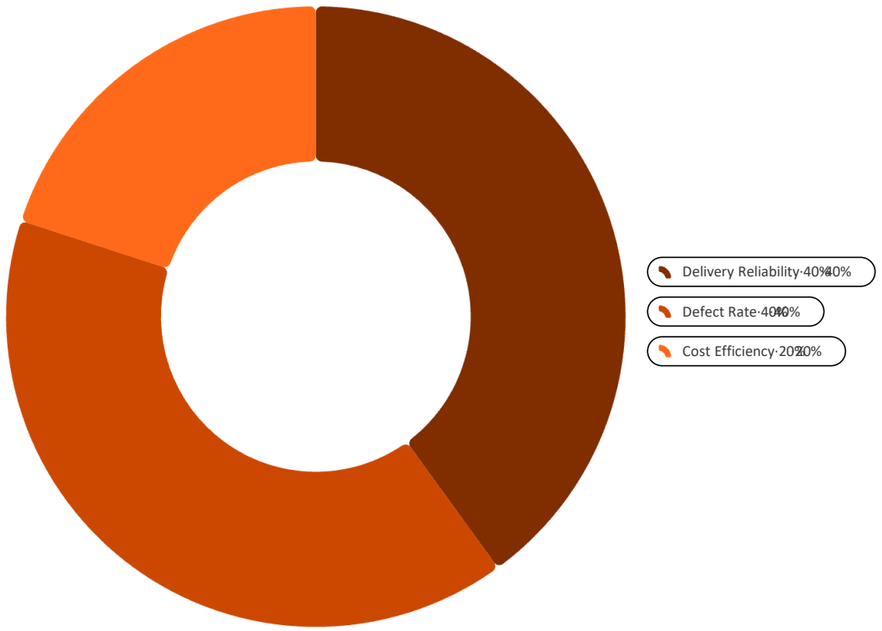
Mean Absolute Error

73K

Training Records

Supplier Performance Scoring

Scoring Breakdown



Automated Vendor Rankings

Every supplier receives a **0–100 composite score**, automatically updated with each transaction. Scores translate directly into letter grades used for procurement decisions.

Grade	Score Range	Status
A+	90–100	Preferred
B	70–89	Approved
C	50–69	Review
F	0–49	Blacklist

Inventory Intelligence Features

Immutable Ledger

Append-only transaction log — every stock movement is permanent and auditable, eliminating data tampering.

Dead Stock Detection

Items with zero movement for **30+ days** are automatically flagged for review before losses escalate.

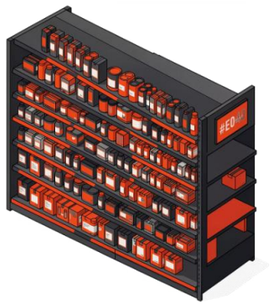
Discount Suggestions

System auto-generates markdown recommendations for flagged dead stock to recover capital quickly.

Stock Rebalancing

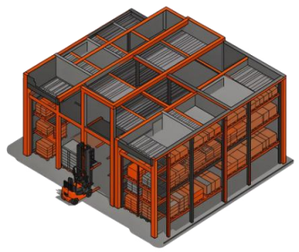
Detects imbalances across warehouses and recommends inter-location transfers to optimise fulfilment.

Real-World Applications



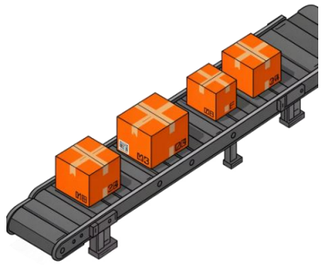
Retail Stores

Prevent shelf gaps and overstock with live replenishment alerts and demand forecasts.



Warehouses

Multi-location tracking and automatic rebalancing keep operations lean and responsive.



E-Commerce

High-velocity SKU management with predictive restocking to avoid lost sales during peak periods.

SMBs

Affordable, deployable alternative to enterprise ERP — designed to scale from day one.

Results & Conclusion

• PROJECT OUTCOMES

~14%

Forecast Error

Mean absolute error on real-world validation data — competitive with commercial solutions.

→ **Eliminated Manual Effort**

Automated stock tracking, scoring, and forecasting replace error-prone spreadsheets entirely.

73K

Records Processed

Real transaction dataset powering the demand forecasting engine.

→ **Smarter Supplier Decisions**

Objective, data-driven ranking removes bias from procurement and vendor management.

3-Layer

Architecture

Production-grade separation of concerns — fully modular and extensible.

→ **Scalable & Production-Ready**

Built with industry patterns (DAO, MVC, Maven) — ready to extend into a full enterprise system.